

Applied Statistics, Fall 2026 Syllabus

MATH-UY 4114 A

Course dates: 09/02/2026 - 12/14/2026

Lectures: Mon & Wed 11:00 AM - 12:20 PM, Jacobs Hall, 6 Metrotech Room 202 (Brooklyn Campus)

Recitation (ARC, Class 9593): Fri 11:00 AM - 12:20 PM, led by Derek Bian

Instructor: Zongyi Li (zongyili@nyu.edu)

Office hours:

Tandon: Mon 2:00 PM - 3:00 PM, room 870, 2 Metrotech

Courant: Tue 2:00 PM - 3:00 PM, room 1027, Warren Weaver Hall

Teaching Assistant: Derek Bian (yb2055@nyu.edu)

Grader: Hongyi Li (hl5452@nyu.edu)

Description: Introduction to the mathematical foundations and techniques of modern statistical analysis. Empirical distributions; Maximum-Likelihood and Bayesian estimators; Theory of unbiased estimators; Statistical decision theory; chi-square, t, and F distributions; Confidence intervals; Hypothesis testing; Linear regression; Bootstrap analysis. If time allowed, we will cover topics in machine learning and statistical learning.

Prerequisites: MA-UY 3014 (Applied Probability) or MA-UY 3514 (Honors Probability). You should be familiar with the main concepts of probability theory such as probability spaces, conditional probabilities, discrete and continuous distributions, random variables, expectation value and variance, cumulative distribution functions, convergence of random variables, laws of large numbers, central limit theorem etc.

Textbook: I will follow my own lecture notes. For further studies, I recommend the following textbooks.

1. Mathematical Statistics and Data Analysis, 3rd Edition, by John Rice
2. Statistical Inference, 2nd ed. by G. Casella and R. L. Berger

Study recommendations: I expect students to study the lecture notes and/or the relevant material in the recommended books between classes. The complexity of the material grows and each class (or book chapter) depends on the previous classes/chapters. Start working on homework assignments early, preferably as soon as they are available. If you have questions, please contact the course staff by email.

Requirements

- **Homework:** Homeworks will be assigned biweekly and constitute 20% of your final grade. All homework must be handwritten. Homework will be assessed for completion and will not be graded for correctness.
- **Quizzes:** A quiz will be given at the end of every Monday class and will constitute 30% of your final grade. The three lowest quiz scores will be dropped; therefore, there will be no make-up quizzes. No make-up quizzes will be offered for ordinary absences.
- **Exams:** There will be a midterm exam and a final exam. The midterm will make 20% of your overall grade and the final exam counts 30%. The midterm and final exam dates are TBA.
- **Grades:** I will follow the standard conversion: A=94-100, A-=90-93, B+=87-89, B=84-86, B-=80-83, C+=77-79, C=74-76, C-=70-73, D+=67-69, D=65-66, F=below 65
- **Make-up policy:** A make-up exam may only be arranged due to documented medical emergencies or a family crisis so that a student has to miss a scheduled exam. Quizzes may not be made up, since the three lowest quiz scores will be dropped. Approved accommodations will be honored in coordination with the Moses Center and the Math Department.

Some Practical Matters

- **Class materials:** All class material will be available via Brightspace.
- **Ed Discussion:** We will use the Ed Discussion forum for all communication and announcements in this class. You are also encouraged to ask questions there and discuss homework problems. You will find a link to Ed Discussion on Brightspace.
- **Gradescope:** We will use Gradescope for homework assignments. You will find a link to Gradescope on Brightspace.

Additional Topics

- **Inclusion:** The NYU Tandon School values an inclusive and equitable environment for all our students. I hope to foster a sense of community in this class and consider it a place where individuals of all backgrounds, beliefs, ethnicities, national origins, gender identities, sexual orientations, religious and political affiliations, and abilities will be treated with respect. It is my intent that all students' learning needs be addressed, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. If this standard is not being upheld, please feel free to speak with me.
- **Student Accessibility:** New York University is committed to providing equal educational opportunity and participation for all students. The [Moses Center for Student Accessibility \(CSA\)](#) works with NYU students to determine and implement appropriate

and reasonable accommodations as well as access available programs and resources to support equal access to a world-class education. Students who are requesting academic accommodations are advised to reach out to the Moses Center as early as possible in the semester for timely assistance.

- **Exam policies:**

- Students will be allowed a maximum of one make-up exam throughout the entire semester per course for an excused reason listed below. There will be no remote makeup exams: All make-up exams must be taken in-person and must be taken within one week of the original exam date.
- If you miss an exam for a medical, religious, or family emergency reason, you must contact the Math Department at soe.math@nyu.edu in order to request a make-up exam within two days of the missed exam. Makeups will not be granted to students who do not notify the Math Department in a timely manner. For medical, religious, or family emergency reasons, in order to qualify for a makeup exam, you would need to be approved for an excused absence for the date of the exam by Tandon Student Affairs. **Excused absence requests must be submitted as soon as possible and no later than 2 days after the original in-class exam date by completing the [form here](#) and must list the exam as a missed assignment.**
- You may request permission for an out-of-sequence exam or final, if you are giving an official presentation or organizing an official session at a suitable out-of-town ACADEMIC conference that directly conflicts with the time of the exam. If you wish to do so, please send an email to soe.math@nyu.edu BEFORE you finalize your travel plans. Permission is not guaranteed. In particular, no accommodation will be made at all for non-academic conferences or for students going to a conference only as an attendee.
- We cannot accommodate out-of-sequence exams, quizzes, and finals for reasons of convenient travel, even if you have already purchased tickets. Please note carefully the date of your exams and final and plan your travel schedule accordingly.
- If you are a student athlete who has an exam conflict for reasons other than practice, please email a letter from NYU Athletics to soe.math@nyu.edu to request a make-up exam at least three days prior to the date of the exam.

- **The Office of Student Advocacy:** At NYU Tandon, we recognize that medical, mental health, or personal issues can sometimes affect your attendance and academic performance. If you are experiencing challenges that you feel are impacting your ability to attend class or succeed in your coursework, we encourage you to reach out to the NYU Tandon Office of Student Advocacy. The Student Advocacy team is here to connect you with the resources and support you need to address these challenges and ensure you can continue to thrive academically. If you begin experiencing any issues that may affect your attendance or academic performance, please reach out to the NYU Tandon Office of Student Advocacy as soon as possible as we will be unable to appropriately address these concerns as the semester progresses, so early communication is essential

to ensure the support and resources you need. The NYU Tandon Office of Student Advocacy can be reached at advocacy.tandonstudentlife@nyu.edu and is located in 5 MetroTech Center LC240.

- **Academic honesty:** The Department of Mathematics reserves the right to impose the strongest academic sanctions for violations of Academic Integrity. Students are responsible for familiarizing themselves with Tandon's Student Code of Conduct including the Policies and Procedures on Academic Misconduct.

Academic dishonesty may include misrepresentation, deception, dishonesty, or any act of falsification committed by a student to influence a grade or other academic evaluation. Academic dishonesty also includes intentionally damaging the academic work of others or assisting other students in acts of dishonesty. Common examples of academically dishonest behavior include, but are not limited to, the following:

- Cheating: intentionally using or attempting to use unauthorized notes, books, electronic media, or electronic communications in an exam; talking with fellow students or looking at another person's work during an exam; submitting work prepared in advance for an in-class examination; having someone take an exam for you or taking an exam for someone else; violating other rules governing the administration of examinations.
- Fabrication: including, but not limited to, falsifying experimental data and/or citations.
- Plagiarism: intentionally or knowingly representing the words or ideas of another as one's own in any academic exercise; failure to attribute direct quotations, paraphrases, or borrowed facts or information.
- Unauthorized collaboration: working together on quizzes and exams that were meant to be done individually.
- Duplicating work: presenting for grading the same work for more than one project or in more than one class, unless express and prior permission has been received from the course instructor(s) or research adviser involved.
- Forgery: altering any academic document, including, but not limited to, academic records, admissions materials, or medical excuses.

Zero tolerance: Any violation of academic integrity will result in a penalty, and the minimum penalty is zero points for the entire exam.

- **Midterm progress reports:** We know timely and actionable feedback is vital for student learning and success. After the midterm exam, you will receive a Midterm Progress Report through NYU Connect. The report will let you know if your progress is strong, satisfactory, or of concern. Strong Progress in this course means you're doing well and are consistently and regularly participating in classroom or online discussions, as well as completing your weekly assignments on time. Satisfactory Progress in this course means you're doing well but there's room to grow and are sometimes participating in classroom or online discussion, as well as completing most of your weekly assignments. Concerns About Progress in this course means your professor has concerns

about your performance, attendance, or other challenges, such as not participating in classroom or online discussions, regularly missing or doing very poorly on submitted assignments, and/or having multiple absences.